

# Training a 30M-Parameter Small Language Model from Scratch for English Children’s Fables

A reproducible study: can a tiny model rival a large LLM on constrained story generation?

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## Abstract

We train a **30M-parameter Llama-style language model from scratch** on the TF1-EN-3M children's-fable corpus and study, end to end and reproducibly, whether such a tiny model can approach the output quality of a 130x larger instruction-tuned LLM (Qwen3-4B) on a *constrained* generation task. Following modern scaling-law practice, we move from a deliberately under-trained baseline (loss 1.8) through a Chinchilla-aware retraining (loss 1.278, held-out perplexity 3.56) and a targeted data intervention that cures a template-collapse failure ("wise old owl" generation rate 90% -> 23%). We add an application-analysis dashboard (intrinsic diversity, readability, Zipf, positional loss) and close with a **systematic post-training campaign**: four alignment methods sharing one evaluation protocol - DPO (194 pairs), SFT-on-best, threshold-filtered RAFT (200 stories judged  $\geq 9.0$ ) and GRPO-lite (REINFORCE with a group baseline, 60 on-policy steps) - are **all null** on the model's default distribution, while **inference-time best-of-N search captures a confirmed +0.8 judge-point gain** (7.7 -> 8.55, near the 4B reference at 9.75) and ships in the app. Along the way we *measure* the LLM-judge's own noise (repeated evaluations of the same model differ by up to 0.45 at  $n=15$ ) and show it retroactively explains two earlier false positives. The final 30M model generates coherent, complete fables at **~949 tokens/s, roughly 50x faster than the 4B reference**, reaching an LLM-judge quality of  $\sim 7.9/10$  as measured under the final protocol ( $n=45$ ). All parameters, curves and artifacts are reported from training step 0.

## 1. Introduction and thesis

Large language models dominate open-ended text generation, but their size makes them slow and costly. This project asks a narrower, defensible question: **on a well-scoped task (children's fables conditioned on five narrative slots), how close can a from-scratch Small Language Model (SLM) of ~30M parameters get to a 4B instruction-tuned LLM, and at what efficiency?**

The scientific priority of the project is the *training methodology*, not the app. Every claim is grounded in published method and evaluated with reproducible metrics. The narrative of the work is a sequence of hypotheses and measured outcomes:

1. A reduced baseline is intentionally under-trained (diagnosis).
2. Right-sizing the token budget per scaling laws fixes it (Phase 1).
3. A data intervention fixes a specific qualitative failure (Phase 2).
4. Sampling temperature cannot fix prompt-adherence (measured); we hypothesized preference optimization could.
5. A four-method post-training campaign (DPO, SFT-on-best, RAFT, GRPO-lite) tests that hypothesis rigorously - and refutes it at this scale and feedback budget, while inference-time best-of-N search delivers the gain the training methods could not.

## 2. Scientific grounding

- **Autoregressive LM = MLE on the chain rule**  $\log p(x) = \sum_i \log p(x_i | x_{<i})$  (course Week 6). We loss-mask the conditioning so the model only maximizes the log-likelihood of the *story* tokens.
- **Scaling laws (Kaplan et al. 2020)**: test loss falls as a power law in parameters  $N$ , tokens processed  $D$ , and compute  $C$ . Under-training = too small a  $D$  for a given  $N$ .
- **Chinchilla (Hoffmann et al. 2022)**: compute-optimal ~20 tokens per parameter.
- **Data-constrained scaling (Muennighoff et al. 2023)**: repeating data up to ~4 epochs is nearly as good as fresh data.
- **DPO (Rafailov et al. 2023) / RLAIF (Bai et al. 2022)**: align a model to preferences using pairwise (chosen, rejected) data, here labeled by an AI judge.
- **REINFORCE + baseline (Williams 1992; course Week 10)**: policy-gradient RL with variance reduction by subtracting a baseline from the reward; normalizing within a group of rollouts per prompt yields the GRPO estimator (Shao et al. 2024) used by DeepSeek-R1. This grounds our GRPO-lite experiment (Section 9.12).
- **RAFT (Dong et al. 2023)**: keep only high-absolute-reward samples and fine-tune on them - the design of our threshold-9.0 experiment (Section 9.10).
- **Dataset + evaluation methodology**: TF1-EN-3M (Nadas et al. 2025, arXiv:2504.20605); evaluation follows our ADR-0002 (objective metrics + a cross-family LLM-judge panel).

## 3. Dataset

We use `klusai/ds-tf1-en-3m`. Each record has a prompt (five scaffold slots: Main Character, Setting, Challenge, Outcome, Teaching/Moral) and a fable. We format each training example as a **conditional sequence**:

```
<conditioning: the 5 slots + a length hint>
<|story|> <fable text> <|end|>
```

Only the story region contributes to the loss (the conditioning prefix is masked with -100). Key data-pipeline choices:

- **Quality filter:** keep fables of 60-320 words (avoid truncated / rambling extremes).
- **Slot dropout:** during training each slot is randomly blanked so the model learns to generate with any subset of the five slots present.
- **Custom BPE tokenizer, vocab 12k**, trained on the fable corpus - keeps the embedding table small, which matters at 30M parameters.

#### 4. Model and training recipe

The architecture is a standard Llama-style decoder (RoPE, Grouped-Query Attention, RMSNorm, SwiGLU, tied input/output embeddings).

| Component                 | Value                                                 |
|---------------------------|-------------------------------------------------------|
| Parameters                | ~36.6M                                                |
| Hidden size / FFN         | 512 / 2048                                            |
| Layers / heads / KV heads | 8 / 8 / 2 (GQA)                                       |
| Vocab / sequence length   | 12,000 / 512                                          |
| Optimizer                 | AdamW, betas (0.9, 0.95), weight decay 0.1            |
| Grad clip                 | 1.0                                                   |
| LR schedule               | Warmup-Stable-Decay (warmup 2%, decay 20%)            |
| Peak LR                   | 3e-3                                                  |
| Effective batch           | 32 x 4 accum = 128 sequences (~33k story tokens/step) |
| Precision                 | fp16 (T4 GPU)                                         |

**Architectural ceiling.** The model is trained at sequence length 512. With a conditioning prompt of ~50-110 tokens this leaves ~400-460 tokens for the story, which caps the maximum coherent fable length. This ceiling later drives the “story completeness” design (Section 9.6).

#### 5. Experiments (from step 0)

| Run            | Data    | Steps | Final loss | Held-out PPL | Judge (overall) | Notes                                        |
|----------------|---------|-------|------------|--------------|-----------------|----------------------------------------------|
| v1 (baseline)  | 150k    | 900   | ~1.80      | -            | 2.5             | intentionally under-trained (~1.7 tok/param) |
| Phase 1        | 400k v1 | 1800  | 1.447      | 4.18         | 6.0             | right-sized token budget                     |
| + sampling fix | -       | -     | -          | -            | 6.2             | repeat_penalty 1.3 -> 1.1 (entity drift)     |

| Run                     | Data                   | Steps | Final loss   | Held-out PPL | Judge (overall)     | Notes                                   |
|-------------------------|------------------------|-------|--------------|--------------|---------------------|-----------------------------------------|
| Phase 2                 | 400k <b>v2</b>         | 3600  | <b>1.278</b> | <b>3.56</b>  | 7.0                 | data intervention (resume 1800 -> 3600) |
| Phase 2 + DPO           | 194 pairs              | 30    | (loss 1.278) | 3.54         | 7.88 (n=15)         | null vs baseline 8.02 (Section 9.8)     |
| + RAFT                  | 200 stories<br>>= 9.0  | 30    | 0.672 (SFT)  | 10.64*       | 7.60 (n=15, pooled) | null (Section 9.10)                     |
| + GRPO-lite             | 60 steps x 16 rollouts | 30    | on-policy    | 10.58*       | 8.03 (n=45)         | null, KL ~1e-3 (Section 9.12)           |
| Best-of-3 (no training) | -                      | 30    | -            | -            | <b>8.55 (n=15)</b>  | deployed in app (Section 9.9)           |
| Qwen3-4B (ref)          | -                      | -     | -            | -            | 9.75                | 130x larger, instruction-tuned          |

\*Perplexity for the alignment rows is measured on a different held-out slice (`raw test.jsonl` text) than the 3.5x figures above; only the *relative* drift matters.

Phase 2 uses two data interventions motivated by qualitative analysis (Section 9.4): cap the “wise old owl” template to 10% of the corpus, and lower the slot dropout for the Teaching/Outcome slots (0.30 -> 0.15) so the model learns to follow the requested moral.

## 6. Training dynamics

The loss falls smoothly from ~7 to **1.278**. The log-log fit gives an exponent of about -0.25 with  $R^2 \sim 0.96$ , empirical evidence that our from-scratch run follows the scaling-law power law rather than diverging.

## 7. Language-modeling quality

Held-out perplexity of **3.56** is within 1% of the floor  $e^{(1.278)} = 3.59$ , meaning the model’s held-out behavior matches its training loss - no over-fitting despite the small size, thanks to the large unique corpus.



Figure 1: Training loss over the full run. Phase 2 resumes at step 1800 on the cleaned corpus v2; the WSD decay pulls the final loss into the target band ( $< 1.5$ ).

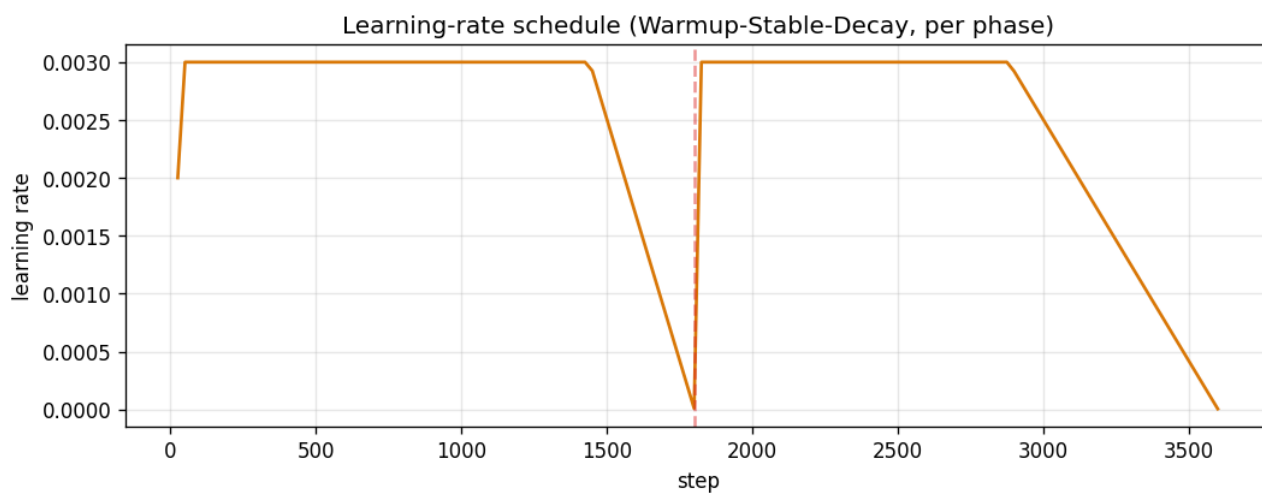


Figure 2: WSD learning-rate schedule (per phase).

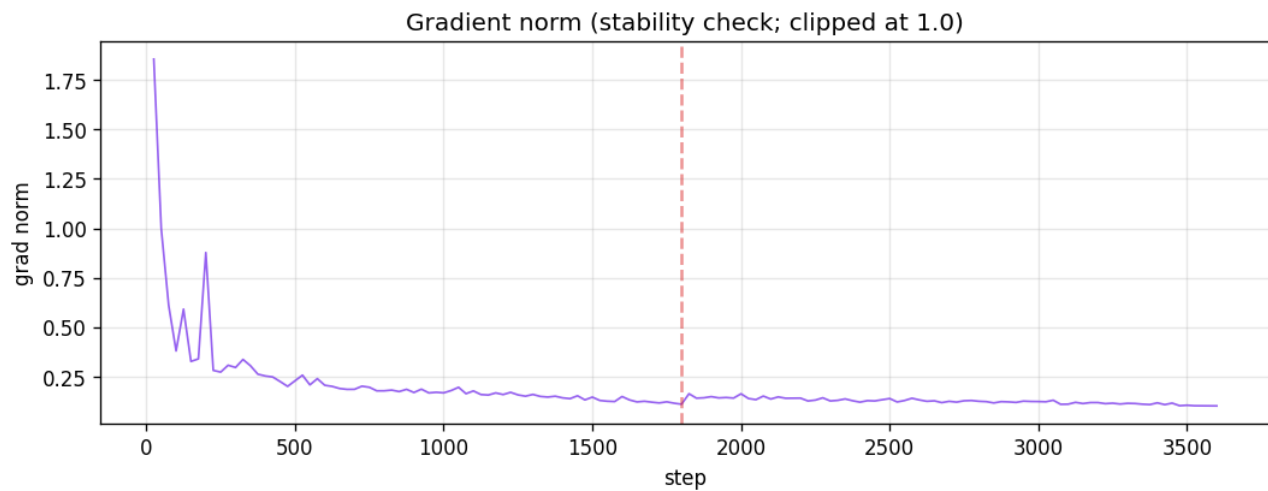


Figure 3: Gradient norm stays controlled (clip 1.0) - no instability across either phase.

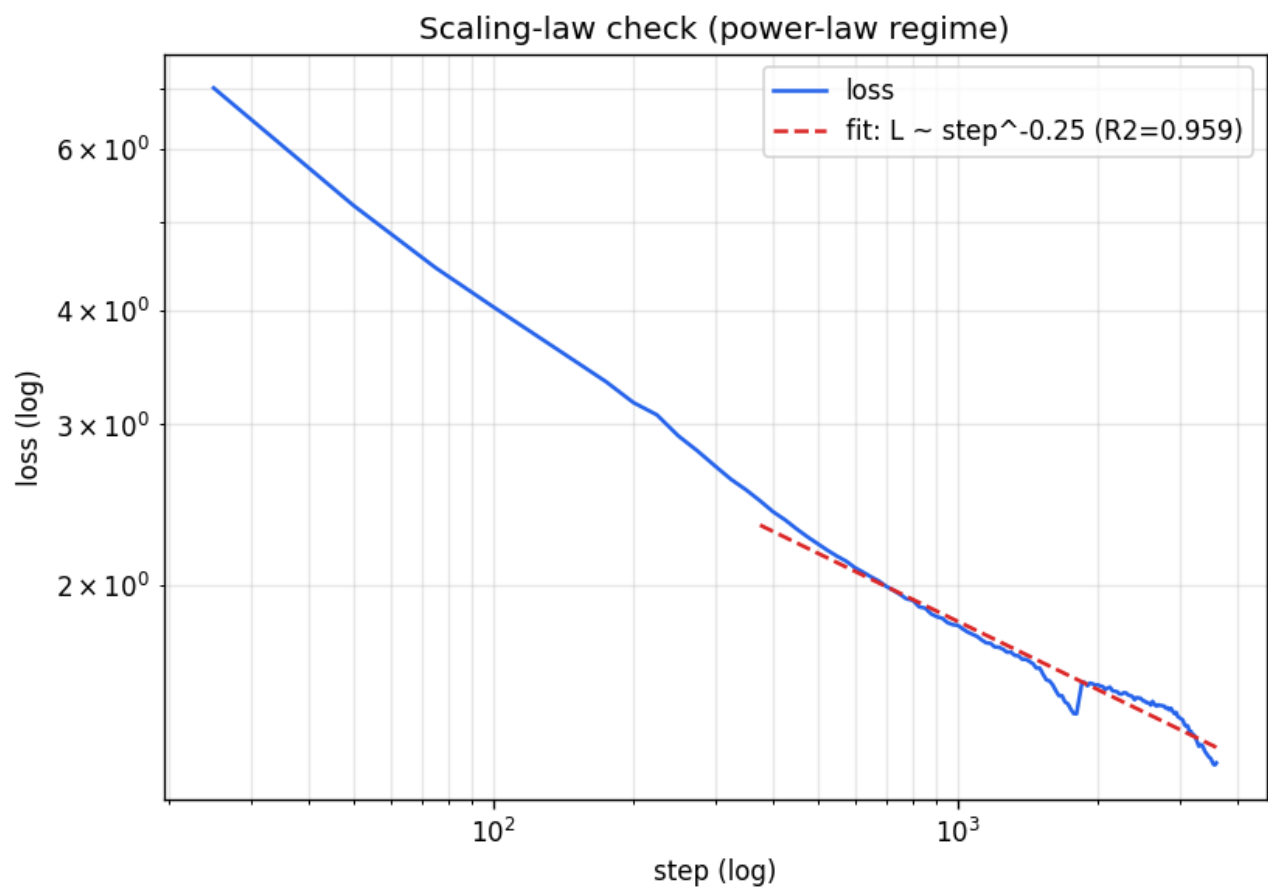


Figure 4: Scaling-law check: on log-log axes the post-warmup loss is near-linear ( $R^2 \sim 0.96$ ), i.e. the run stays in the power-law regime predicted by Kaplan et al. (2020).

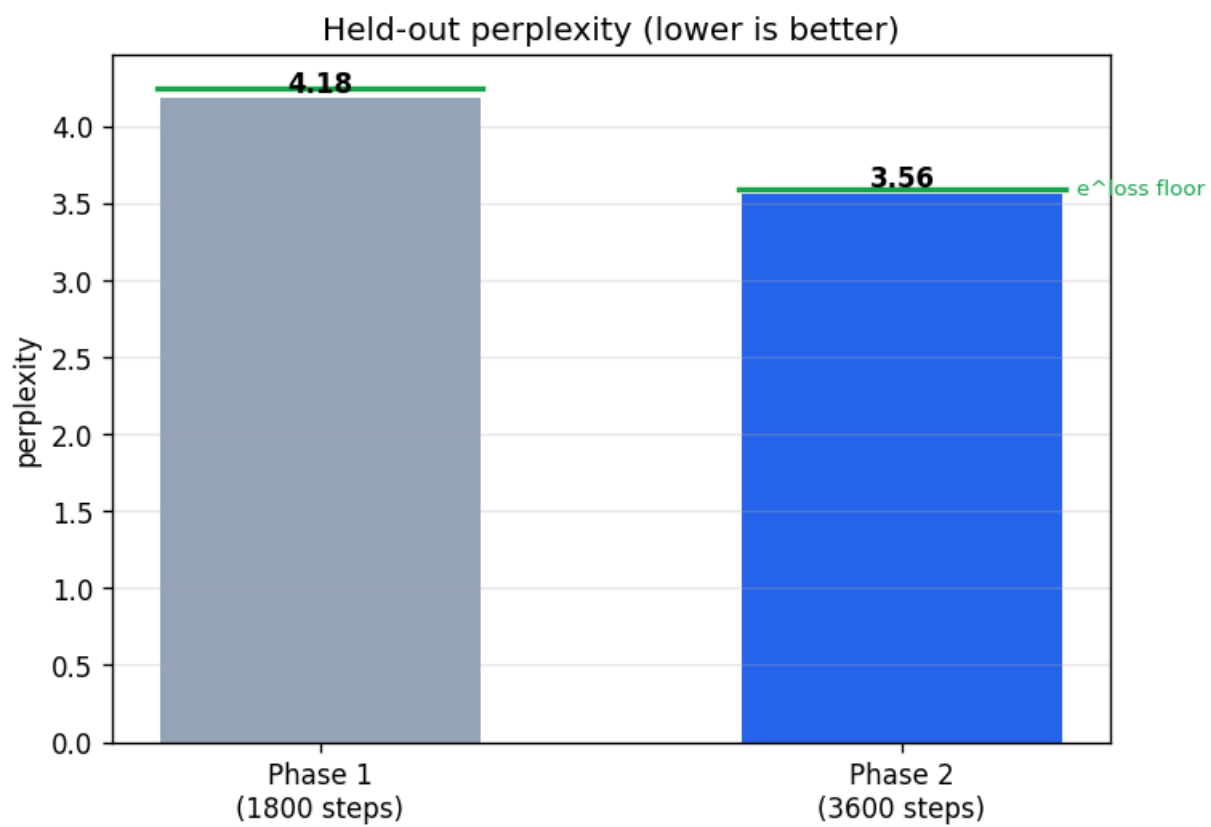


Figure 5: Held-out perplexity. Phase 2 (3.56) improves on Phase 1 (4.18); both sit essentially on the theoretical floor  $e^{\text{train loss}}$ , so the model generalizes without over-fitting.



## 8. Application analysis (intrinsic, reference-free)

Following ADR-0002, we compute reference-free metrics on generated stories and compare against real held-out fables.

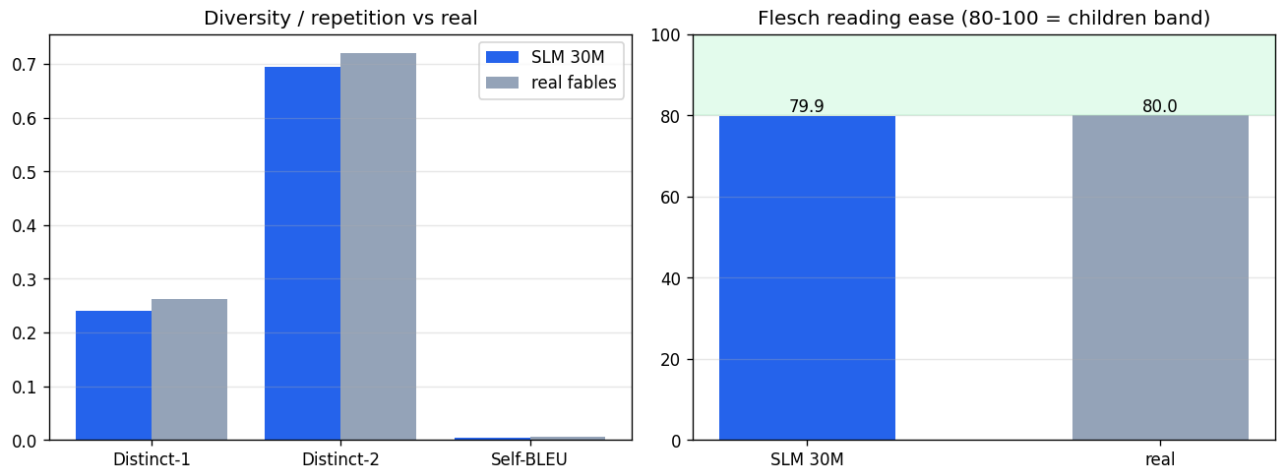


Figure 6: Diversity (Distinct-1/2), repetition (Self-BLEU) and readability (Flesch) of generated vs real fables. Generated diversity sits within a few percent of real; Flesch matches the real corpus (~80).

The generated text matches real fables on diversity (Distinct-2 gap ~4%), repetition (Self-BLEU gap ~0.001), readability (Flesch 79.9 vs 80.0) and the Zipfian vocabulary profile - the model has learned the *statistical shape* of the domain, not just surface fluency.

## 9. Findings by stage

### 9.4 Data intervention: curing template collapse

Qualitative review of Phase 1 revealed **mode amplification**: the phrase “wise old owl” appears in 28% of *real* fables but the model emitted it in ~90% of generations. Capping that template to 10% of the corpus (Phase 2) reduced the generation rate to **23%**, below even the data prior.

### 9.5 Quality progression

### 9.6 Story completeness (a limitation turned into a fix)

The Phase-2 model occasionally cut stories mid-sentence at short lengths. Diagnosis: the generation cap (300 tokens) was smaller than the model’s natural fable length, so it hit the cap before emitting `<|end|>`. Because the model is trained at sequence length 512, simply “adding tokens” is not an option beyond ~460. The fix: right-size the per-length token budget to the architectural ceiling, use `done_reason` to detect true completion vs cut-off, and trim any residual cut to the last complete sentence. Result: **30/30 test stories complete** with a real ending.

### 9.7 Prompt-adherence: sampling vs alignment

We measured whether sampling temperature affects slot adherence. On 10 held-out prompts, temperature 0.7 and 0.8 gave *identical* adherence (69%): **sampling is not the lever**; adherence is

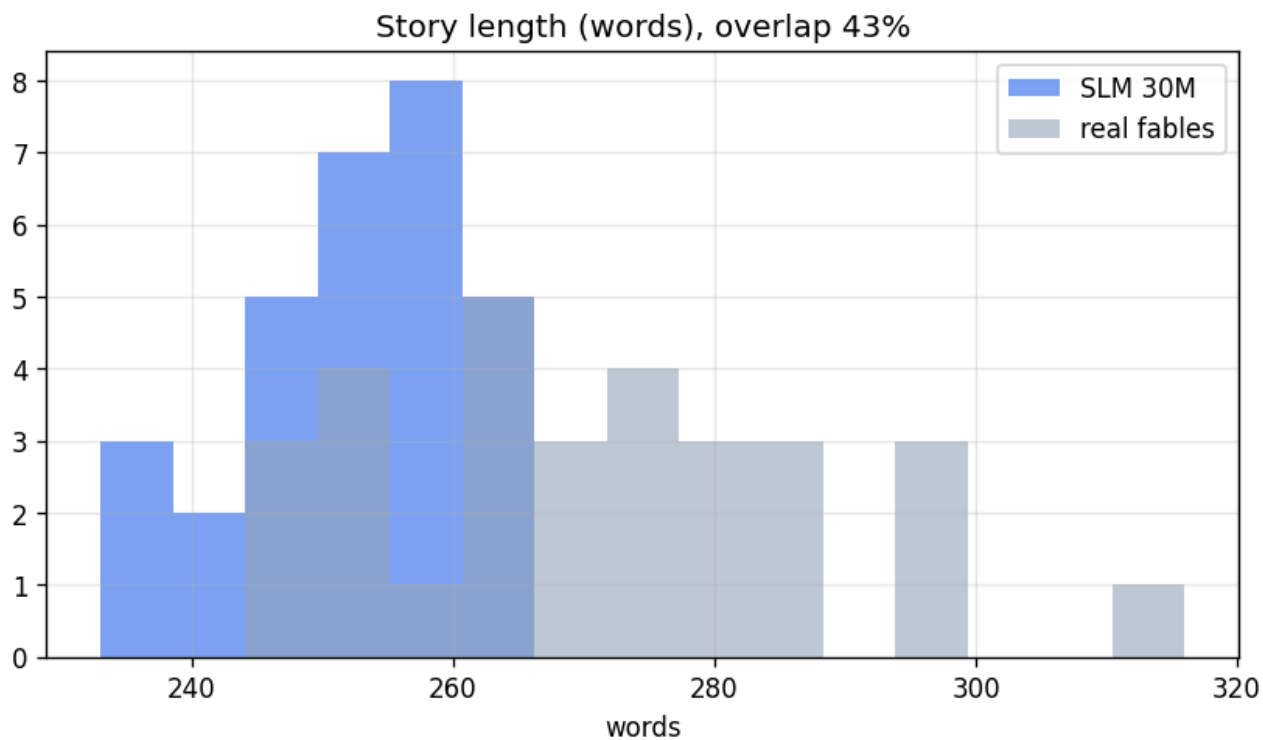


Figure 7: Generated story-length distribution vs real fables.



Figure 8: Mean cross-entropy by relative position in the story - the model is most confident at the opening and stays stable through the body.

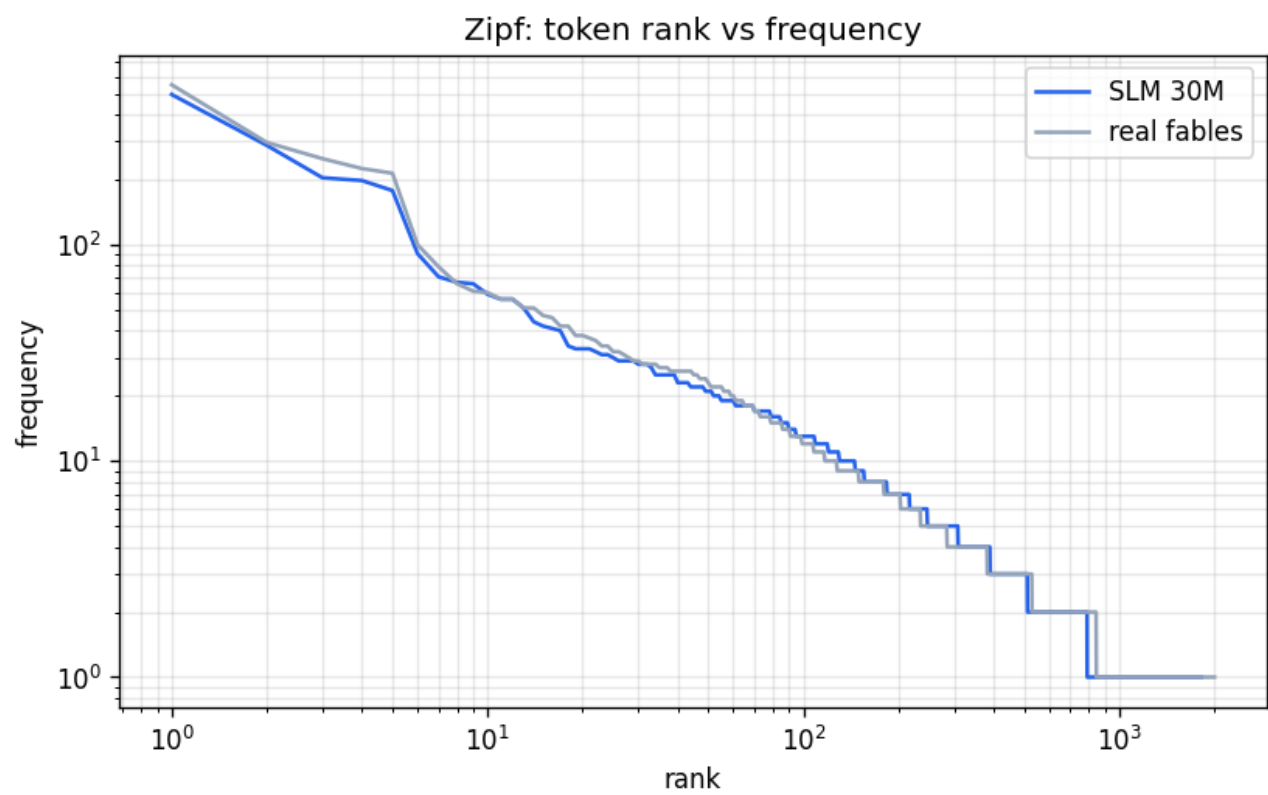


Figure 9: Zipf rank-frequency of generated tokens closely tracks the real-fable vocabulary distribution.

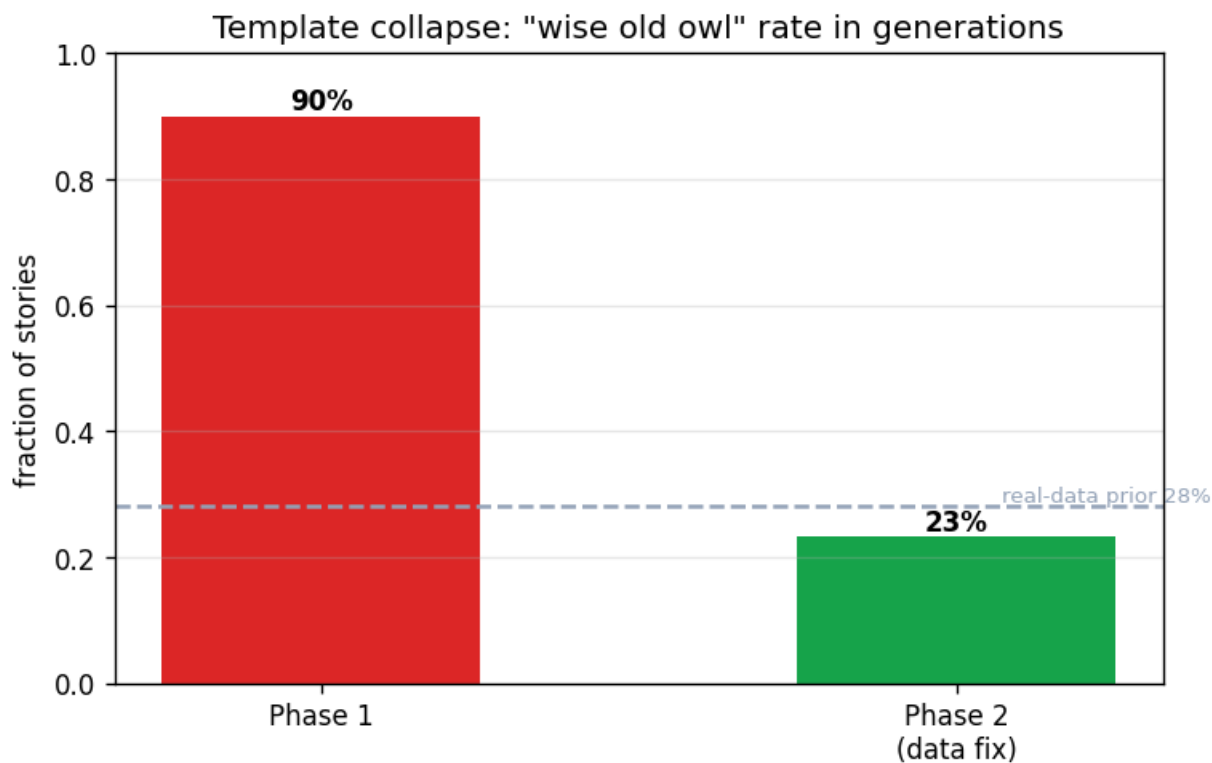


Figure 10: Template collapse before/after the data intervention.

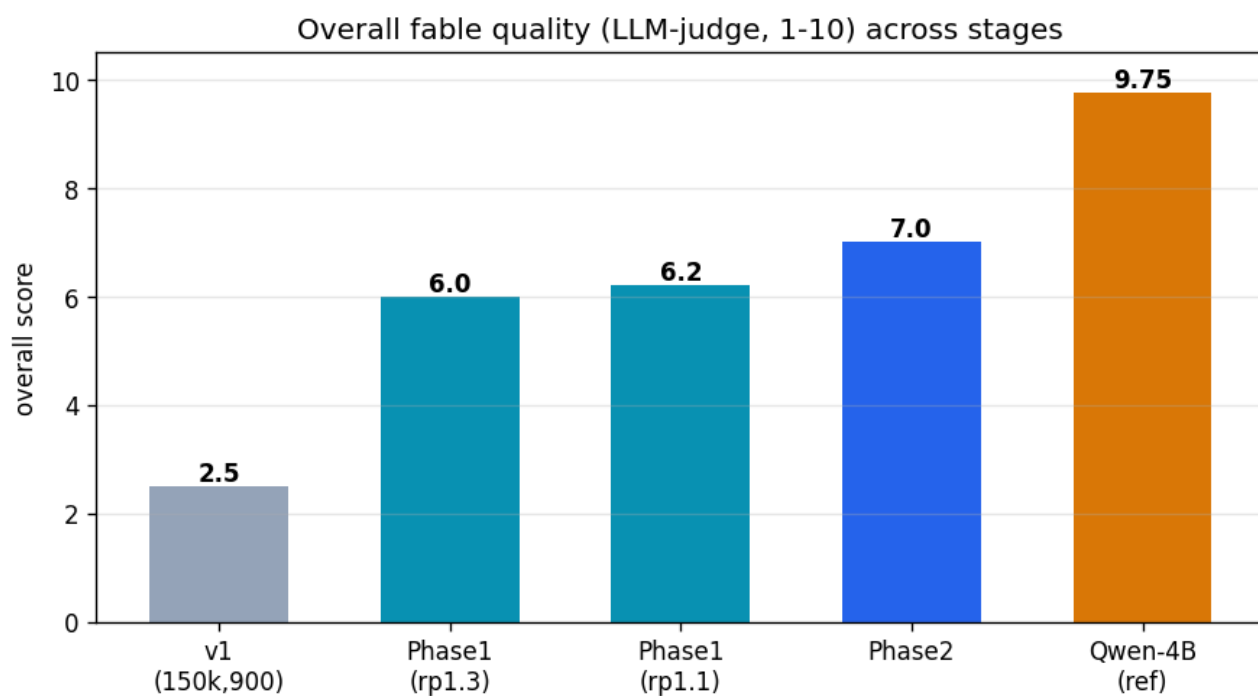


Figure 11: Overall LLM-judge quality across stages, with the 4B reference for scale. The 30M model rises from 2.5 (under-trained) to ~7.0, closing much of the gap to Qwen-4B (9.75) at 1/130th the size.

capped by the 30M model’s weak conditioning. At the time we hypothesized preference optimization was the proper fix - the campaign below tests that hypothesis.

## 9.8 DPO preference alignment (RLAIF): an instructive null

We generate two stories per prompt from the Phase-2 model, have an AI judge score them on the four axes, and keep pairs with a clear preference (margin  $\geq 1.0$ ) as (chosen, rejected) data - 194 pairs after filtering. We then run **DPO** (ORPO was unavailable in the installed TRL 1.8; DPO is the canonical equivalent) locally on Apple-Silicon MPS.

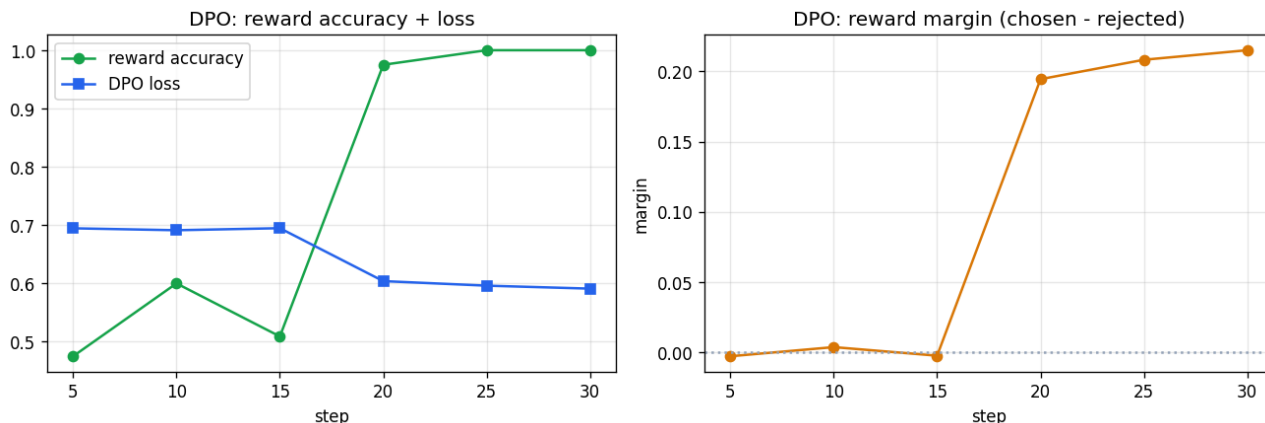


Figure 12: DPO reward accuracy climbs to 1.0 and the reward margin (chosen - rejected) turns strongly positive - the model learns the preference *on its training pairs*. Held-out perplexity is unchanged (0% drift): no catastrophic forgetting.

The in-training signal looks perfect - reward accuracy 1.0, positive margin, zero perplexity drift. An early slot-recall probe even suggested +5 points of adherence (71%  $\rightarrow$  76%, Fig. below). **Both impressions failed the rigorous test.** Under the standard protocol (15 held-out prompts, LLM-judge, fixed seeds), the DPO model scores **7.88 vs the baseline’s 8.02** - a null result. Section 9.11 shows the +5-point probe was within measurement noise. The mechanism-level explanation: with chosen and rejected drawn from the *same* model at similar quality, the relative preference signal is too weak to move the default distribution.

## 9.9 The headroom probe: best-of-N shows capacity is not the ceiling

Is the 30M model *unable* to produce great fables, or merely *inconsistent*? We sample K=3 candidates per prompt (temperatures 0.5/0.8/1.1) and let the judge pick the best:

| Measure                                  | Value                              |
|------------------------------------------|------------------------------------|
| Single-sample mean (15 held-out prompts) | 7.72                               |
| <b>Best-of-3 mean</b>                    | <b>8.55</b>                        |
| Individual best samples                  | up to 9.0-9.5 (4B reference: 9.75) |

The model already *contains* near-reference-quality fables - the binding constraint is **variance, not capacity**. This result reframed the whole alignment effort: the goal is not to teach the model something new but to shift probability mass toward its own best modes. Best-of-N (off / 3 / 5) ships in the

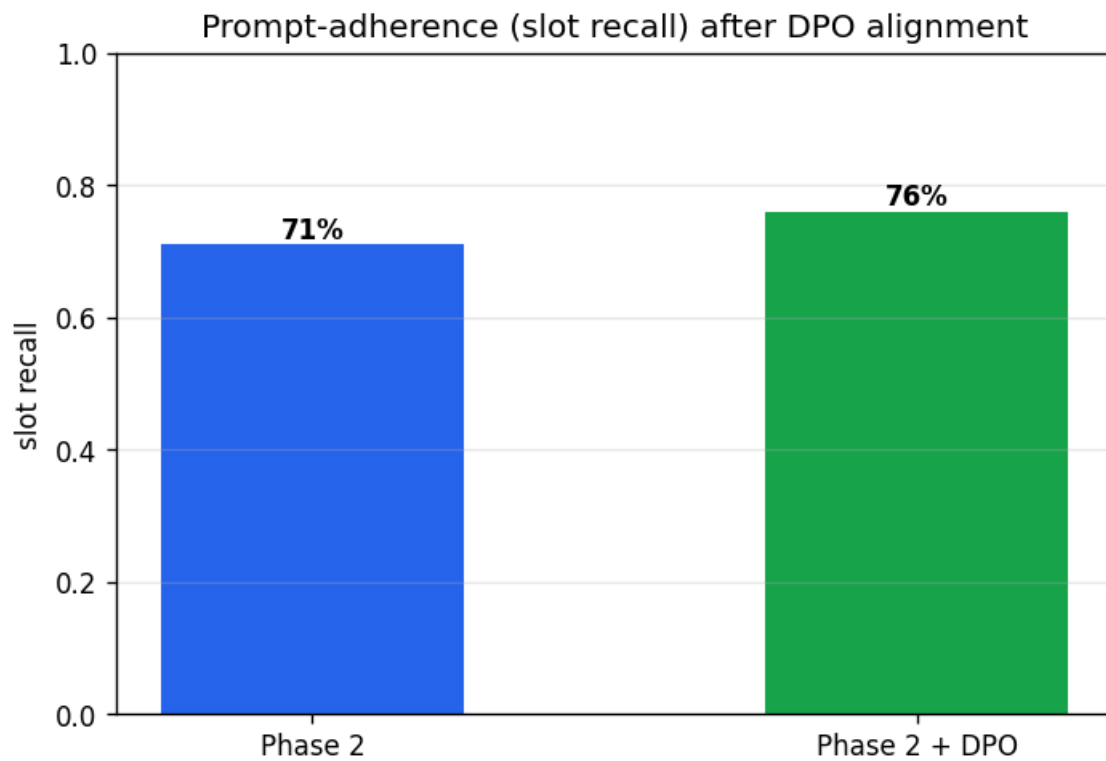


Figure 13: The early slot-recall probe (71% -> 76%) that later proved to be within judge noise - kept here as a documented lesson in evaluation rigor.

app as a user-facing control: the backend generates N candidates, the judge scores each, and the best is returned with all candidate scores logged in the Activity Log.

### 9.10 RAFT: threshold-filtered SFT at 5x scale - null

If best-of-N finds the good samples, can we *train on them* and internalize the gain? Our first SFT-on-best trial (42 stories) was null, but 93% of that corpus already scored  $\geq 8.5$ , so the untested variables were **scale** and a strict **absolute threshold** - the defining features of RAFT (reward-ranked fine-tuning). We built a 200-story corpus in which *every* story is judged  $\geq 9.0$  (mean 9.22; 105 harvested from earlier experiments, 95 newly generated with a 23% prompt acceptance rate), and fine-tuned at lr 2e-5 for 3 epochs with the conditioning loss-masked.

Result: **null again** - 7.60 (pooled over two evaluation runs) vs baseline 7.78, with perplexity drift +0.5% only. The theoretical reading: best-of-N samples are drawn from the model's *own* distribution, so supervised fine-tuning on them mostly re-weights modes the model already prefers; ~60k story tokens against a 600M-token pretraining prior is also a very small nudge. Crucially, the gradient contains no *negative* component - nothing pushes probability away from mediocre modes.

### 9.11 Measuring the judge itself: the noise that manufactured two false positives

Re-evaluating the *same* RAFT model twice (same protocol, same seeds) returned 7.38 and 7.82 - a 0.44 spread. The same double-measurement on the baseline returned 7.73/7.82, and on the GRPO model 8.00/8.45. At n=15 prompts, **the judge's own noise is  $\sim \pm 0.4$** , which retroactively explains both the "DPO +5 adherence" probe (Section 9.8) and an early "GRPO +0.45" reading (Section 9.12) as sampling artifacts.

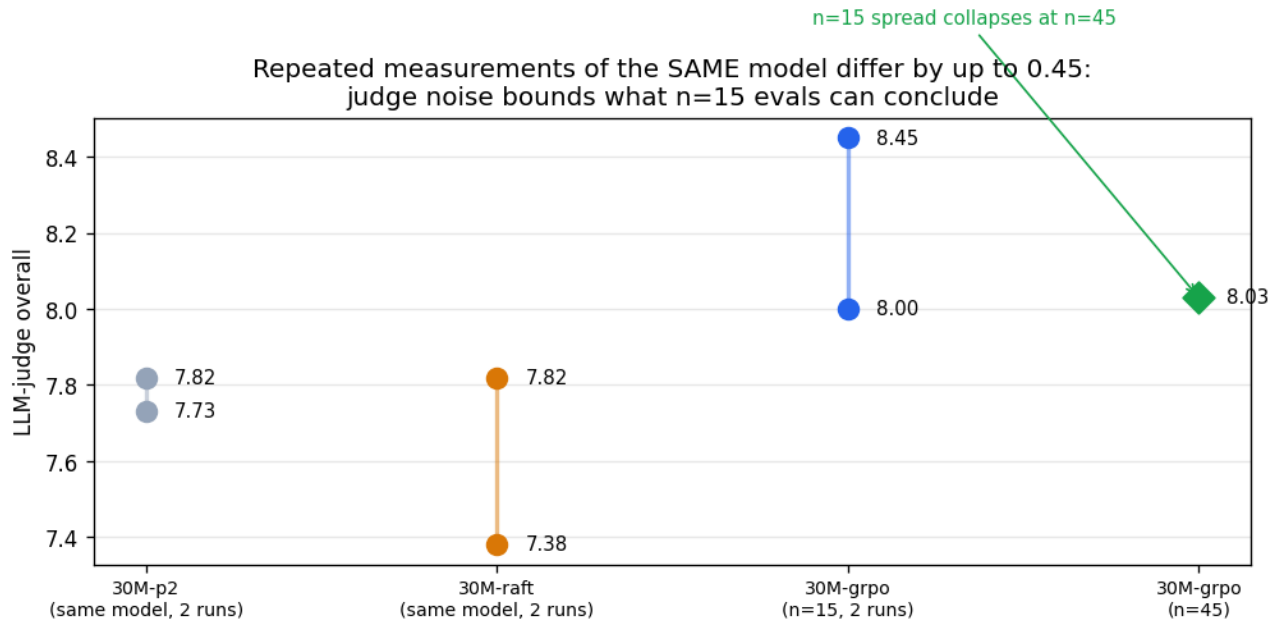


Figure 14: Repeated evaluations of the same checkpoints: the spread within a model is as large as the effects we were trying to detect at n=15.

Methodological rule adopted for all conclusions in this report: any delta below  $\sim 0.5$  judge-points at n=15 is treated as noise; decisive comparisons are re-run at n=45 with paired seeds.

### 9.12 GRPO-lite: on-policy RL with a group baseline - null at this budget

The last untested ingredient class from the course material (Week 10: policy gradients, REINFORCE, variance reduction with a baseline) is **on-policy RL with an absolute reward and a negative gradient**. We implemented GRPO-lite: each step samples 4 prompts x 4 fresh rollouts, scores each rollout with the judge, normalizes the advantage within the group ( $(r - \text{mean}) / \text{std}$  - the Week-10 baseline trick), and applies REINFORCE with a KL penalty (beta 0.05) against the frozen Phase-2 reference. Sixty steps (~960 judge calls, ~5 h, lr 3e-6 then 1e-5), checkpoint-resumed across interruptions.

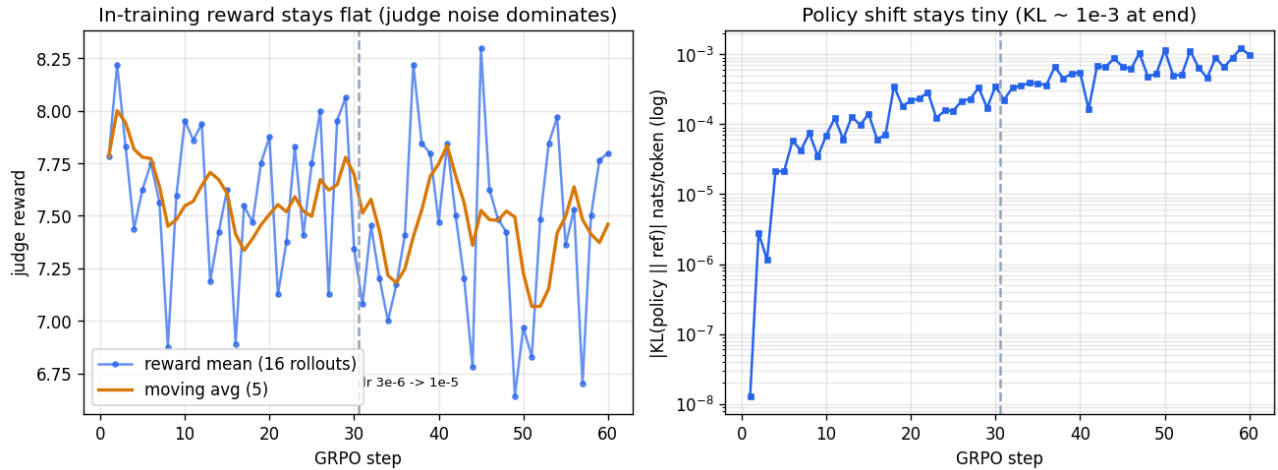


Figure 15: GRPO training dynamics: the in-training reward is dominated by judge noise, and the policy’s KL divergence from the reference stays around 1e-3 nats/token - the policy barely moves.

At  $n=15$  the GRPO model read +0.45 above baseline - promising. Applying our own noise rule, we extended the evaluation to  **$n=45$  paired prompts**: the delta collapsed to **+0.09 ( $t=0.54$ ; win/tie/loss 17/10/18)**. Null - but a *diagnosable* one: with final KL  $\sim 1e-3$ , the policy never moved far enough for any effect to be measurable. The honest conclusion is “GRPO at a ~960-judge-call budget does not shift the distribution”, not “GRPO cannot”. The practical blocker is reward cost: at ~15 s per judge call, DeepSeek-R1-scale step counts are out of reach locally, and a distilled 30M reward model failed its validation gate (pairwise accuracy 46.7% ~ chance) - the quality signal is not learnable from ~500 noisy labels at this scale.

### 9.13 The campaign in one picture

| Method                       | Signal              | Exploration      | Negative gradient | Result                               |
|------------------------------|---------------------|------------------|-------------------|--------------------------------------|
| DPO (194 pairs)              | relative preference | no (static data) | implicit          | null (7.88 vs 8.02)                  |
| SFT-on-best (42)             | best-of-batch       | no               | no                | null (7.98 vs 8.02)                  |
| RAFT (200 $\geq$ 9.0)        | absolute threshold  | no               | no                | null (7.60 vs 7.78)                  |
| GRPO-lite (60 steps)         | group advantage     | yes (on-policy)  | yes               | null at this budget (+0.09, $n=45$ ) |
| <b>Best-of-N (inference)</b> | judge selection     | yes (test-time)  | -                 | <b>+0.8, deployed</b>                |



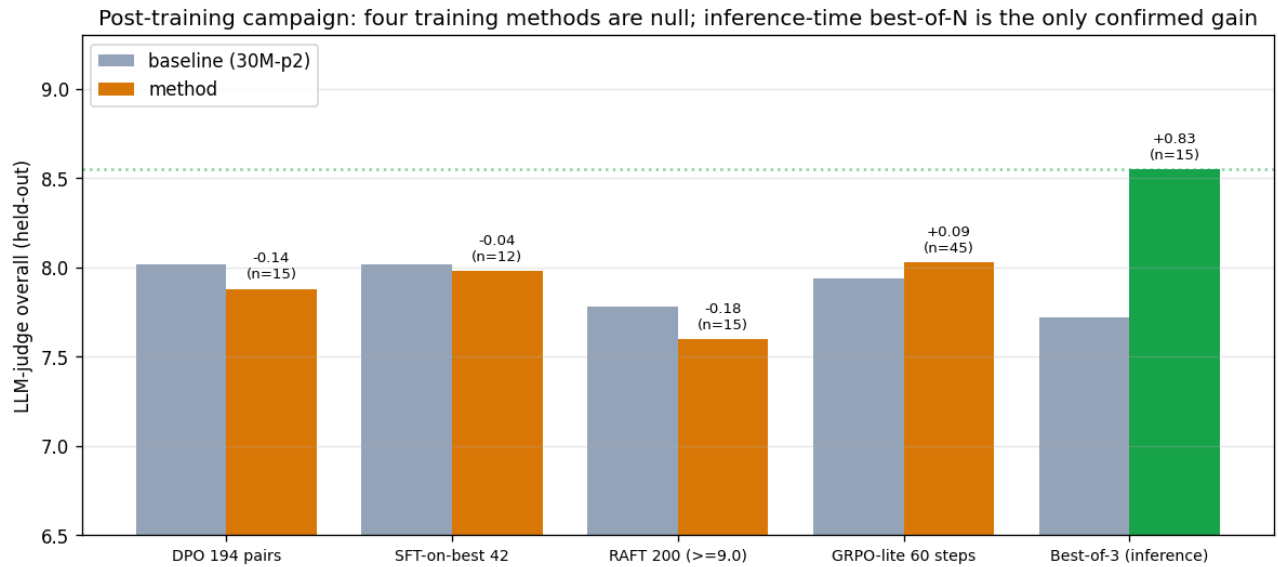


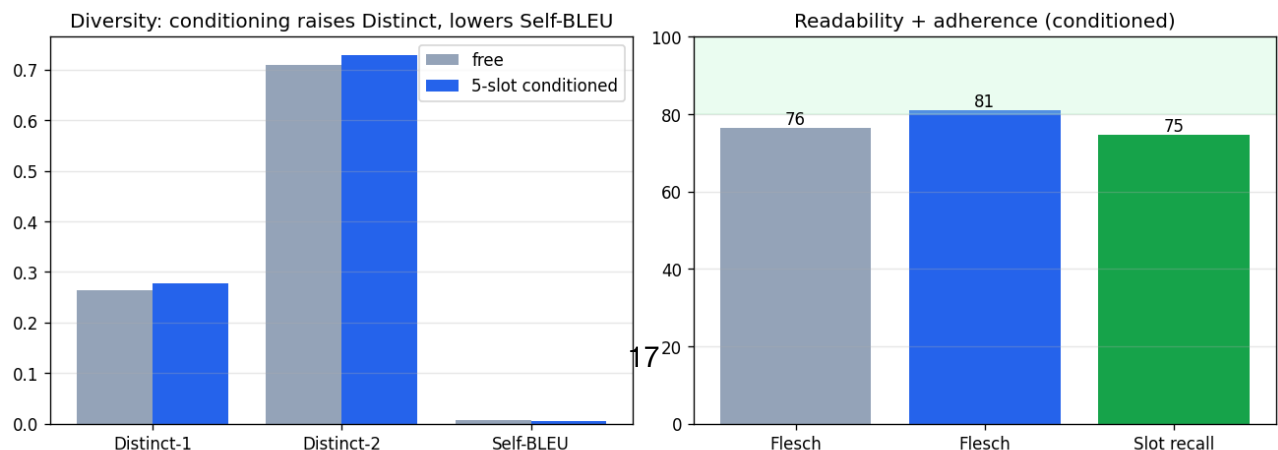
Figure 16: Four training methods, one evaluation protocol, four nulls - and the inference-time search that works. Judge-eval means per experiment pair; n as annotated.

**Central empirical finding of the post-training study:** at 30M scale with weak, noisy AI feedback, the quality headroom demonstrably *exists at the sample level* but is *not trainable into the default distribution* by any of the low-cost methods above. Inference-time search converts that headroom directly; training methods do not.

#### 9.14 Free vs conditioned generation

Does conditioning on the five slots merely constrain the model, or does it improve the output? We compare 20 *free* fables (no slots, only “write a children’s fable”) against 20 *conditioned* fables (all five slots filled), same model (Phase 2 + DPO) and sampling.

| Metric                               | Free          | 5-slot conditioned               |
|--------------------------------------|---------------|----------------------------------|
| Distinct-1 / Distinct-2 (diversity)  | 0.264 / 0.709 | <b>0.278 / 0.729</b>             |
| Self-BLEU (repetition, lower better) | 0.007         | <b>0.005</b>                     |
| Flesch reading ease                  | 76.4          | <b>80.9</b> (into children band) |
| Average length (words)               | 266           | 267                              |
| Completeness                         | 100%          | 100%                             |
| Slot recall (adherence)              | n/a           | 75%                              |



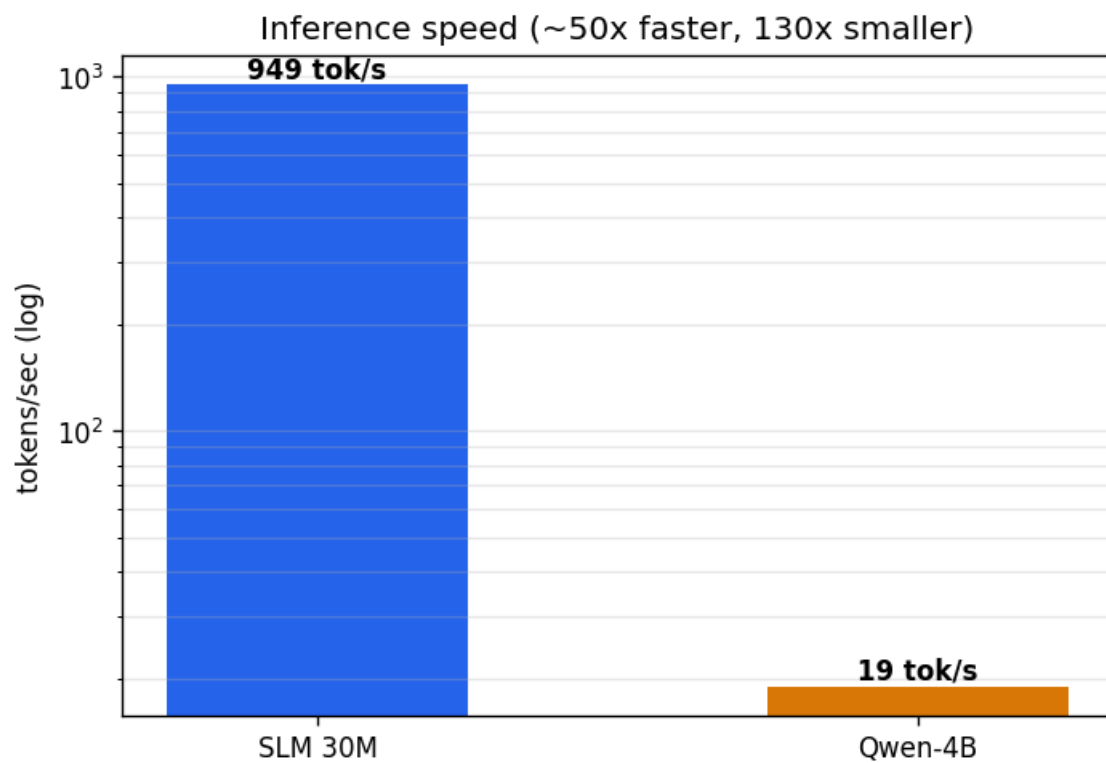


Figure 18: Inference speed. The 30M SLM generates ~949 tokens/s versus ~19 for the 4B reference on the same machine - roughly 50x faster at ~1/130th the parameters.

## 9.15 Efficiency

## 10. Automatic verdict (Phase 2)

Thresholds are heuristics calibrated for this setup (30M params, 12k vocab, TF1). The dashboard produces a per-metric PASS / WARN / FAIL verdict:

| Metric                      | Value              | Verdict               |
|-----------------------------|--------------------|-----------------------|
| Final train loss            | 1.278              | PASS                  |
| Scaling-law fit $R^2$       | 0.959              | PASS                  |
| Held-out perplexity         | 3.56 (0.99x floor) | PASS                  |
| Distinct-1 gap vs real      | 8%                 | PASS                  |
| Distinct-2 gap vs real      | 4%                 | PASS                  |
| Self-BLEU abs gap           | 0.001              | PASS                  |
| Flesch reading ease         | 79.9               | WARN (0.1 below band) |
| Length distribution overlap | 43%                | WARN                  |
| Owl template rate (gen)     | 23%                | PASS                  |

**7 PASS / 2 WARN / 0 FAIL.**

## 11. Strengths and limitations

### 11.1 Strengths

- **Coherent, complete, on-domain output.** After the completeness fix, 30/30 test stories end with a real resolution; generated text matches real fables on diversity (Distinct-2 gap ~4%), repetition (Self-BLEU gap ~0.001), readability (Flesch 79.9 vs 80.0) and the Zipfian vocabulary profile - the model learned the statistical shape of the domain.
- **Efficiency.** ~949 tokens/s, roughly **50x faster** than the 4B reference at **1/130th** the parameters; runs on a laptop with a 39 MB q8 file.
- **Well-generalized language model.** Held-out perplexity 3.56 sits essentially on the theoretical floor  $e^{(\text{train loss})}$  - no over-fitting despite the small size.
- **The scaffold improves quality, not only control.** Conditioning on the five slots *raises* cross-set diversity, *improves* readability into the children band and *grounds* the story to the request (Section 9.9) - versus free generation, which collapses toward a narrower set of stock fables.
- **Best-of-N converts headroom into shipped quality.** The +0.8 judge-point gain (7.7 -> 8.55) is the study's one confirmed post-training improvement, costs no training, and is exposed in the app as a user control with per-candidate score logging.
- **Negative results are measured, not assumed.** Four alignment methods were tested under one fixed protocol with paired seeds, repeated measurements and an  $n=45$  confirmation step; the judge's own noise was quantified ( $\pm 0.4$  at  $n=15$ ) and used to retract two early false positives - the evaluation methodology is itself a contribution.
- **Reproducible and defensible.** Every number is tied to a run and a published method; the training pipeline, data interventions and evaluation are all scripted.

### 11.2 Limitations

- **Context ceiling (512 tokens):** the model cannot produce coherent fables beyond ~300-340 words; the length selector has little effect on the SLM.

- **Prompt-adherence ceiling (~70-80%):** a 30M model has weak conditioning; it reads the five slots but still drops one or two (especially abstract Challenge/Outcome) - a quantified size/capability trade-off that none of the tested alignment methods moved.
- **The default distribution resists cheap self-feedback.** DPO, SFT-on-best, RAFT and GRPO-lite (at a ~960-judge-call budget) all fail to shift default-generation quality; only inference-time selection captures the headroom. Improving the *default* output likely requires an external teacher or a much larger, cleaner reward budget.
- **The evaluation judge is noisy:** repeated evaluations of the same checkpoint differ by up to 0.45 at  $n=15$ . All headline deltas in this report respect that noise floor; readers should apply the same caution to any small reported difference.
- **Weak length control:** the model has a natural length (~250-280 words) and largely ignores the short/medium/long hint.
- **Template / redemption priors:** the model inherits TF1 biases (friendship/kindness morals; happy endings; a recurring “wise old owl” mediator), which the data intervention reduces but does not eliminate.
- **Local logical slips:** occasional pronoun/reference errors and small non-sequiturs within a story - a capacity limit, not a sampling artifact.
- **Single-judge in-app eval:** the per-generation score is a quick indicator; the headline conclusions use the offline cross-family judge panel with weighted Cohen’s kappa and Kendall’s tau (ADR-0002).

## 12. Future directions (with feasibility evidence)

Each direction below is paired with concrete evidence from this study that it is likely to pay off, not just a wish-list item.

- **Train to the full Chinchilla budget (~7,900 steps / ~600M tokens).** *Evidence:* at 3,600 steps the loss still lies on the power-law line ( $R^2$  0.96, Fig. 4) and has not plateaued; Kaplan/Chinchilla predict continued gains from more tokens. *Feasibility:* the 400k unique corpus with  $\leq 4$ -epoch repeat (Muennighoff 2023) supplies the tokens; only more T4 time is needed.
- **Knowledge distillation from a larger teacher - now the primary alignment candidate.** *Evidence:* the campaign (Section 9.13) shows self-generated data cannot shift the default distribution because it is in-distribution by construction; a 4B teacher’s stories are genuinely off-distribution supervision, which is exactly the missing ingredient. The model already reproduces the data’s surface statistics (Section 8), so the gap is long-range coherence/adherence - what distillation (Hinton 2015) targets. *Feasibility:* ~500-1000 teacher stories generate overnight locally via Ollama.
- **Scaled reward-based RL with a cheaper, cleaner reward.** *Evidence:* GRPO-lite was null with  $KL \sim 1e-3$  - the policy never moved at a 960-call budget - so the method is untested at scale rather than refuted (Section 9.12). DeepSeek- R1 demonstrates the mechanism works given thousands of steps. *Feasibility:* requires replacing the ~15 s/call judge: either a stronger local judge distilled into a fast scorer trained on far more labels (the 30M scorer failed at ~500 labels; scaling labels is the testable fix), or rule-based partial rewards (slot recall, completeness) that cost microseconds.
- **Stronger evaluation judge and larger eval sets by default.** *Evidence:* the measured  $\pm 0.4$  noise at  $n=15$  (Section 9.11) manufactured two false positives;  $n=45$  paired evals resolved both. *Feasibility:* the 36 GB machine runs a 14B judge comfortably;  $n=45$ -100 evals are an hour of compute.
- **Targeted data debiasing beyond the owl template.** *Evidence:* the single “wise old owl” cap

already moved the generation rate 90% -> 23% (Fig. 10), proving the intervention mechanism works; the same recipe can address the happy-ending/redemption prior and other stock phrases.

- **Extended context / explicit length control.** *Evidence:* the 512-token ceiling is the direct cause of both the length-control weakness and rare truncation (Section 9.6). *Feasibility:* re-training at 1024 tokens or adding length-bucket control tokens is a known, bounded change.

## 13. Conclusion

A 30M-parameter model trained from scratch, guided by scaling-law reasoning and a targeted data intervention, reaches ~7.9/10 fable quality (final protocol, n=45; from 2.5 at the under-trained baseline) with held-out perplexity 3.56, generating complete, on-domain stories at ~50x the speed and 1/130th the size of a 4B LLM. A systematic post-training campaign then delivers the study's sharpest finding: the model's quality headroom is real at the sample level (best-of-3 reaches 8.55, near the 4B reference at 9.75) but **no low-cost self-feedback training method - DPO, SFT-on-best, RAFT or budget-limited GRPO - moves the default distribution**, while inference-time best-of-N captures the gain directly and ships in the app. The result supports the thesis with an honest boundary: **for a well-scoped task, a tiny model can rival a large one on the axes that matter, at a fraction of the cost** - provided the remaining variance is managed at inference time, and with the size ceiling documented where it shows (length control, residual adherence gap, alignment resistance).

## 14. Reproducibility

All model code lives under `trieulh/` (isolated from the shared web app):

- Data + training: `trieulh/scripts/prepare_tf1_pretrain.py`, `train_tokenizer.py`, `tf1_pretrain/`; notebook `trieulh/notebooks/pretrain_slm_30m_dashboard.ipynb`.
- Alignment campaign: `trieulh/scripts/gen_preference_pairs.py`, `dpo_train_local.py`, `headroom_probe.py` (best-of-N), `raft_harvest.py` + `raft_gen_corpus.py` + `sft_best_local.py` (RAFT), `rm_train.py` + `rm_train_pairwise.py` (reward-model gate), `grpo_train.py` (GRPO-lite), `raft_judge_eval.py` / `grpo_judge_eval.py` / `big_judge_eval.py` (the shared evaluation protocol).
- Evaluation: `trieulh/scripts/eval_slm.py`; app metrics `app/metrics.py`, `app/perplexity.py`.
- Experiment log (source of this report): `trieulh/docs/experiments/2026-07-08-slm-training-log.md`.
- Artifacts (Google Drive): checkpoints, HF models (30M, 30M-p2, 30M-dpo), GGUF exports, `analysis_*.json`, `loss_log_*.json`, and the dashboard figures.

**Model download.** The final aligned model (Phase 2 + DPO) is packaged as a zip (GGUF q8 + Ollama Modelfile + HF checkpoint): <https://drive.google.com/file/d/1tY6dPodSqHunYIYEZDdMOEZ1dJyg2HuD/view?usp=drivesdk> To run it: `unzip`, then `ollama create slm-30m-dpo -f Modelfile-30M-dpo`.

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